Active Inference LiveStream 056.2 Neural Generative Coding, Active Infere

Livestream | *2:07:30*

All right. Hello and welcome everyone. It's November 21st, 2023. We're here in ActInf livestream number 56.2 discussing a quartet of papers with Alex Arurbia. This is the third discussion we've had. We had a .0 giving some background and context.
Great. Hello and welcome everyone. It's November 21st, 2022. We're in ActInf livestream.
Empirical demonstration of the feedback delay of livestreaming. So let's just jump in. Perhaps Blue, if you have any initial kind of remarks or questions, then to Alex. And also if anyone's watching live, they can of course write a question and I'll check it out. So go for it, Blue, and we'll just jump in. So for me today, I think I'm really excited to get into the cognitive model. So like the last paper was my favorite and it was I think maybe like the most zoomed out. So I'm really kind of looking forward to getting into that and then maybe exploring what that would look like like in a non-human system. So that's I mean, I kind of always want to go there, but I'm still I still want to go there.
Cool. What figure should we jump to or what part of the paper should we jump to, Alex?
Alex. I mean, whichever one caught, I think yours and Blue's interest, I guess we could go to the architecture one. Oh, I do want to, if you don't mind, a tiny shameless plug that I just want to take the moment right now. I'm going to post this link in the chat because it relates to our conversation we had in point one. So this is a paper that just came out the end of last week.
I just want to add that as contextualization, especially Blue, you might find it interesting because this was work I worked on. I think I even alluded to it over the like the last seven months. This is why I was talking about inactive and embodied cognition. And there's a lot more. So now, now it's out there. I don't have to pretend like, oh, it's a secret project. That was it. So that's my only shameless plug. Let's go back. I should start with this. So the, the, um, the as not to steal the thunders of 56.1 were this paper that came out in the time between. Yeah. It literally came out last end of last week. Yeah. Wow. So, I mean, maybe just while we're here, like, could you just, you don't have to review it. I've already had some, you already have so many papers, but it might be worth it. Yeah. Like, could you just give us like the elevator pitch about this paper and, um, like maybe your favorite figure? Cause I mean, we were really excited for your comments last week.
Okay. So scroll up, Daniel. Uh, it's, it's actually early. There's, there's cool stuff, but go to the, it's a very long paper. Yeah. That picture. It has, it also has an appendix or supplement, but, uh, so, and it was a longer original paper too. Uh, just that way, again, I don't steal a thunder of this particular podcast cause our focus was different. Um, now the, the general idea is, uh, my, my thesis or Carl and my Cthist thesis is about a moral computation, which was a phrase that Jeffrey Hinton, uh, introduced, uh, very briefly, like in about like two sentences or a small paragraph in his paper in December. Uh, and I have written at length about it in a couple of my own papers, end of December of last year and January and February. Um, so this is sort of like, uh, the big,

uh, deep dive into what does it mean to make a computer that is not immortal, which is what we have in computer science today. Uh, the computer is inextricably bound or rather the software is inextricably bound to the hardware. Uh, sorry, not bound to the hardware. I'm just making a mistake there. So if for example, we can copy our software or machine intelligence program across any amount of computers and it doesn't really matter. Um, and, uh, it's also extremely energy inefficient. And then the opposite side of that is mortal computation, which is essentially drawing inspiration from life itself. Uh, for example, uh, when any one of us dies, a human dies, uh, your computations, your memories, your quirks, the things that define who you are, are essentially gone, right? Because it's unique to you and that is because you're an embodied inactive agent. Um, and so what's kind of nice about this diagram is it's, it's a really colorful, pretty diagram, but it's in, it sort of synthesizes two out of the three perspectives in this paper. So, uh, Carl and I investigated biophysics, uh, which is sort of like the intersection of physics, physiology, biology, chemistry, and the ideas that have echoed over, I mean, essentially the last century. Uh, and that's the top half. The bottom half is a classical and very interesting domain called cybernetics, which is like the intersection of, uh, computer science, control theory, also physics, and a bunch of other domains and information theory. And it also has similar ideas and everything centers around this idea of homeostasis and all the millions of, uh, variations and higher order, uh, you know, uh, concepts that build on that. And the ideas that, uh, our intelligence and, uh, essentially is driven by our, our per desire to persist and to survive in an environment. And so this actually takes my comments from the last point one podcast with you guys, uh, to its natural conclusion, which is, I don't just believe you need a body, you need an environment, you need, essentially a drive to persist and to fight, uh, thermodynamic dissolution or dissipation into the environment or disintegration is a better word. Um, so we are essentially applying the biological second law of thermodynamics, which is why it's also about free energy. Uh, and so then, uh, we, and then there's another person, but it's also, that it's a later diagram, cognitive science. And, uh, Carl and I, uh, propose 5E cognitive theory, which is just taking the classical 4E theory. Um, I don't remember if I said this to you last time, but embedded, uh, extended, uh, inactive, and, uh, you know, the, uh, yeah, embedded, oh, embodied, the 4E, the traditional 4Es, we added something called elementary cognition, which I did say to you guys, basal cognition last time. So this kind of like ties these three areas together, as well as the philosophy of life and death and existentialism and kinds of combines them together into a unified framework. And then, uh, we, we dive into Markov blankets. Uh, I, I think they're a beautiful and elegant formalism for tying these ideas together and getting us a step towards, uh, you know, the mortal computation, mortal computer design. And then we, we, we talk about the free energy principle underpinnings that essentially the, the mortal, uh, computer embodies. And we review, uh, a couple of existing mortal computers, which do exist. So, uh, organoids or microphysiological intelligence is a beautiful example of this. Uh, even, uh, we talk about fungal computers, also Michael Levin's work on xenobots. So we have like a nice little table and then there's an appendix with a little bit of expansion on that. Uh, also we have to give credit to, uh, Ashby, uh, who's a really well-known and, or, uh, a very important cybernetical scientist,

cyber cybernetic, cybernetician, um, and his homeostat, which is also an example of an in-silicon
moral computer. So long story short, uh, we, if we want one path to artificial general intelligence
is not going to be the way we have it now. Uh, it's not through just bigger and bigger transformers,
which are extremely energy efficient. They're labeled in the domain of AI itself, red AI. If we
want green AI, we want edge computing, we want neuro robots, uh, that are also able to do human-like
actions and, uh, enact in human-like behavior, uh, because we think for example, the human model or
even animal models, right, are a good template for how we can interact and engage in our environment.
Um, then the idea is that we need to transition to this idea of mortal computation and leave behind,
my last point here, uh, the tool oriented view of artificial general intelligence, because the way
we treat AGI today, and we say this towards the end of the paper, um, that while it's very important
and very helpful for humanity, uh, to build artificial intelligence systems that are very, very narrow and
very tool-centric, they are more to support something for a particular task. If we want to build general
intelligence, uh, then, uh, Carl and I put forth the mortal computation thesis as one possible pathway,
uh, to go into the future. So this is sort of also a call for researchers across all these different
domains to come together, ranging from biophysics to biology to computer science and machine learning,
to computational neuroscience, cognitive science, it's pretty much, and cybernetics, of course,
all of them kind of converging, perhaps build, uh, a different pathway to what we call the mortal
computer. And so I think that wasn't quite a good elevator pitch. That was a little bit longer, but
I'm going to stop there because I think then you can just read the paper. It's, it's, it's fun.
Well, it's a tall elevator. There's a lot of floors in the bunker.
I think there's a perfect articulation there. 56.2, let's focus on living, functioning, cognitive
frameworks, but it's really cool that you're going into the life and death elements that way. So,
I do think that the paper that Blue wants to talk about, there's a compliment to it because we do
mention very briefly in this paper, cognitive architectures, like the common, and also the
common model of cognition blueprint that we also talked about in point one, which is the center basis
for the paper that, uh, Blue was going to ask some questions about. I think, uh, it's mentioned here
briefly in that we also talked about as that being, uh, a potential, maybe we don't say it this way.
I said it in conversation to someone, but a, uh, a car, an inductive bias, which you did write,
uh, you wrote basis. I also talk about it as like the idea of cognitive architecture and modules of mind
serve sort of like as a prior, and that is, uh, produced by evolution. And, uh, we also talk in the
paper too, the moral computing paper evolution is important. Natural selection is an important
ingredient. Uh, and if we don't use that, uh, then we have to design the results of evolution. So,
I think the common model of cognition, which can help segue into the paper now, um, is a beautiful bias.
It is one of our best bets for how do we emulate the modules of mind and how things interact. Um,
and this would get at the, the neural computation component of a mortal computer. Obviously,
it does not say anything about the body, about the niche, about other agents, but, you know, you can imagine
it is a little block that would go into this framework and that would be one step towards a
mortal computer. So. So can I just temporarily derail us? I just have to ask a follow-up question.

I have to ask. I fall for derailing at the beginning. Um, so just in terms when you're,
when you're talking about mortal computing, I mean, I just couldn't help but go to quantum computing,
where the computation itself is so heavily dependent on the substrate and the modality of the hardware,
right? Like the, the computation, the hardware and the computation are like really inextricably linked
there. Um, and, and in the same way that like, you know, I can write a paper, you can write a paper,
all of these things will outlast us. Like, so some of our computations will survive beyond like the
point where we're biologically dead. Right? So in a quantum computer, similarly, all of those computations can
then be taken and translated and put into Python and all, all of the things. Right. Um, but I, I just
wonder to what, um, extent, and like, I know, I, I kind of know where quantum computers and artificial
intelligence like meet, you know, like everything's gotta go into the cubo and then the answer comes out.
And so then we take the answer and put it back into Python and do regular AI things with it or in some
language, right? Not always Python, but sometimes, right? So where do you think can a quantum computer
or can we ever get quantum computation or what is the role of quantum computation? How can that
intermingle with artificial intelligence and then become a mortal computer? Or is that just like
too far fetched? We're not going to think about that.
Well, so the paper doesn't go into that specifically. However, we touch on a little bit because, uh,
a little bit of the paper and we, we cited plenty, uh, Carl's work with Chris Fields, which I know he's
been on, you guys have been doing a series with him on quantum computing, quantum information theory.
Um, he had a paper, uh, neuromorphic computing induces the free energy principle or some version
of that title. I read that many times as well. And in there I think is the way that quantum computing
or quantum information theory would interact with the mortal computer. And so I don't see,
I see it as another perspective. There could have been a fourth perspective in this already long paper.
Um, and, uh, I did originally have some ambition to write something about that. And we distilled it
to just the concept that Chris Fields introduces, which was the interface. So I think the quantum
information theoretic interpretation of the brain and the body, uh, especially like as presented in that
paper and related work could be a better way to deal with the Markov blanket formalism that, uh, Carl and I,
uh, constructed in this paper, this paper, uh, which is largely grounded by also my educational background too,
because I only have, uh, you know, some moderate knowledge in, you know, quantum mechanics. Uh, and it
was a while since I studied that, but, um, in there we are treating the paper is like a classical
interpretation, or you could say almost the Newtonian interpretation of the free energy principle.
And, uh, and Carl and I discussed this many times as we were working on this paper, um, that, that has a
limitation. It's not necessarily that you can't build a more. I think this paper gives you all the
foundation you need to literally build a mortal computer or get started. It's a, like a theoretical
mathematical conceptualization. However, um, there are some weird things about the thermodynamic
energy exchange between the internal states of your agent, your entity, and the environment. And
originally I had a different diagram of the Markov blanket so that there is later in the paper,

you'll see it. And I really liked that diagram because it really made it. Yeah. You probably saw it. It was there towards the end. It looked like a spider almost. Um, yeah, that. So this originally had a different version where I had these, uh, extra states, uh, because I like to take old concepts and extend them myself. And I had these thermodynamic states in there and I had them tried and I had probabilistic arcs that wired them. All these arcs are designed carefully to make sure that it adheres to the Markov blanket principles of like, you know, the dependency relationships between children and parents and the children of children or the children of parents, um, and how they relate together. So I had embedded extra states and Carl said that I should remove those because it's, uh, it violates the classical interpretation of the free energy principle. And long story short is that you have this energy exchange in the diagram and you can represent that as dissipation, but it's not a state that the agent would have like access to in the Markov blanket. So now back to the Chris Fields interpretation. If you go to the interface, uh, I think he calls it like a quantum interface. He added me or, um, he had a name for it. It has like this really fancy symbol in the, in the paper. Um, if you use that, you get access, it's not like you have access to manipulate it, but you can sort of perceive it from the perspective of the agent. You see the sensory and the action states or the, uh, the partitions of that field. Um, and I'm probably butchering his language, Chris's language, but, and then you also have like the thermodynamic exchange portion or partition of that field. Uh, and like, it's like, uh, the non-manipulatable and manipulable portions of it. And so I think that interpretation could be a more powerful way to, like you said, blue earlier that quantum information theory and quantum, uh, sorry, quantum information, theoretic and quantum computing interpretations lend themselves to that natural entanglement that this, that this paper, uh, takes from. So again, we surveyed work all the way back to Descartes, uh, and Emmanuel Kahn, but you know, and tying all their ideas together. Cause by the way, even though we're putting forth the phrase mortal computation, the mortal computation thesis, uh, and trying to essentially establish that as our foundation for all of science to build on, uh, it's kind of been echoed, right? Uh, brilliant minds, thinkers, theoreticians, engineers have existed forever. So, uh, it's kind of like, uh, a synthesis and unification of already wonderful ideas across vast parts of all the domains. Um, but I think the quantum information theoretic way could be the next step, the fourth perspective to quantum computing, and it could lend itself to maybe easier implementation, especially if you have access to a quantum computer. Obviously this is the joke we had last time, uh, when we talked about evolution and why I specifically might not directly work in that. Although I have some work in, uh, genetic algorithms and neuroevolutionary with, uh, Travis Dessel, who was on here for the ant colony optimization work that was presented. Um, it's compute, right? I don't have access to the quantum computers of our time or maybe the simulator. And of course the knowledge to perhaps use that effectively. However, I think that someone who does could perhaps take the principles that are put forth here, take the Chris Fields, uh, uh, interface interpretation. And he goes into a lot more detail. Like he talks about, he had something called like a T-D-N-N or something. It was like the quantum neural network and they were trying to come up with the morphology. But the, uh, the principles there are echoed in some sense here too. And we cite that word, but, um, obviously, but, uh, Carl's on there too, but, uh, the, the, uh, the principles of morphology, uh, which are like essentially, and we originally had another word in the title called morphic, uh, because the idea is that intelligence should ultimately, you can't divorce it from any form of instantiation. You got to choose something. You're going to be brittle, but you're going to be thermodynamically efficient. And I think if you could interpret it from that perspective, the quantum information theoretic that might lend itself to a more powerful implementation, uh, the last comment I'll make, and I'm already long-winded on this. So the final point is we do also briefly mention in the conclusion of this paper, you know, we haven't even talked about the papers that, uh, we wanted to about today, but in the conclusion, uh, I also mentioned quantum computing, but in a weird way. So I hope it doesn't feel too contradictory to that. I just said that if you take the quantum information, theoretic interpretation of mortal computing or build your own interpretation, because we don't really do it much for you here, uh, except mention the, the, the interface, um, while that could be a powerful and perhaps the best way, who knows, to implement it, we also mentioned quantum computation as sort of like, okay, others have been looking at this in AI. And for example, saying like we can do, if we have a properly developed, uh, quantum computer, we can do big brute force search, right? We can do simulated annealing faster than our minds ever could have conceived with classical computing. And that's kind of just been in what's we can do machine learning better. Um, and we, we sort of argue a little tiny bit against it, uh, not against the interpretation, but rather the, the fact that quantum computing plus machine learning is just another way to do what we currently do, which is narrow AI brute force search better and faster. And so that's essentially, it keeps alive immortal computation, right? Because what are we going to probably do? I, I, I, I not, uh, not pretending to be a seer. I'm not pretending to predict the future. There's no way I can know of all people, but, um, the way we're going about it is that, oh, we're going to try to put chat GPT or some transformer like model in a quantum computer. And yeah, it's going to work a lot faster. It might be more efficient, but at the end of the day, we're going to be programming that quantum computer at a separation level, right? We're going to try, try to deal with it. Like we do in a normal operating system. Maybe we all have to learn a little quantum mechanics to do it. Um, and we write our code and we write our software, and then we just hope that the quantum computer can execute it really fast. That's not the premise that I would want a quantum computer to be used. That's still immortal computation just done in an accelerated faster way. So we do warn or rather are rather even just more like slightly common that while quantum computing will be important and, uh, we, there is, uh, promise there to even speed up the computations we do today. It's not necessarily its current use, or the way we sort of project to use it is not the right path to AGI either. So, or biomimetic intelligence, which is really what the paper's trying to say. If we want to build biomimetic AGI, maybe there's other types of AGI. And I don't mean like animal, because animals are definitely a part of this paper, uh, organisms, anything that maintains homeostasis to some level. That's what this paper says is the, the pathway to that style of AGI. There might be something else. Again, I don't want to comment on whether or not, uh, large language models and transformers and these new architectures, there might be something alien that I can't foresee. And maybe quantum computing will make that, uh, you know, more energy efficient in some way or faster. It's still going to carry forward though, the premise of tool-based AI and, uh, you know, AI, uh, in the way that we do a narrow AI. I don't think it's going to lend itself to the path that we would really want general intelligence and general intelligence really is tied, uh, at least if we use organisms as our exemplar, it's tied to the fact that we, we don't have forever, right? Uh, that we, we know to some degree, whether we're conscious or unconsciously aware of the fact that I am at odds with my environment, I'm going to behave in this way in service of that. And of course, evolution, uh, encode some of that knowledge going into it. Um, so I just want to comment that I, I think that it's, it depends on how you use the quantum perspective. Like if you use it from like the Chris Fields quantum information, theoretic interpretation of the markup blanket, that might be a really excellent way for someone to even follow up on this and add the fourth perspective that Carl and I kind of just dance around a little bit or an implementation. I think there could be some really fascinating things here, but if you're just using quantum computing, uh, to accelerate your normal run, it's current day machine learning, the current approaches that we have, uh, then you're just still carrying forward the banner of immortal computation. And again, too, I want to make one last comment just because we talked about it. Um, this paper says we need mortal computation and, uh, it was weird. There isn't, uh, much more than Jeff's one mention of it in his 2022 forward forward learning paper. Then my paper had a little bit longer section, the predictive forward forward algorithm. And then I had the spiking version of the forward forward, uh, the, the pioneered that idea. And we had like an introduction, like about a page about mortal computation. And then there's this one with Carl, which is, you know, a really, really long, uh, doozy of a paper. Um, so aside from that, I've only seen mortal computation sort of discussed, Oh, I see a little bit of mortal combat appearing. Um, I see discussions like brief mentions of mortal computation, but there was something that Jeff mentioned somewhere along the line, or maybe it was an interpretation of it. I don't also think mortal computation completely replace immortal computation. I think both have a place in society. We just need to understand that they both do different things and they both have different purposes. If we want, you know, intelligent agents and neuro robots, I'm on the camp. You need mortal computation to do it, right? But, uh, if we want that type of general behavior, but immortal computation is not going to go away. It's just more like we understand it has a very high thermodynamic cost because we have to deal with the von Neumann bottleneck, the memory bottleneck. We're not using a memory processing. Um, maybe quantum computing will speed that up too. And then immortal computation can live side by side and we'll have those as tools. So I also want to make that clear for anyone that ever watches, you know, this recording later, uh, that the mortal computation thesis isn't to say it's the only way to do computation. It's just, we should shake up the roots of computer science at its foundation and say, maybe there's another way to do it and we should have both live side by side and Hey, this one might lend itself to AGI. Um, which by the way, some people might disagree with that as well. I don't, again, I, we do offer it very humbly in the paper as one possible pathway, not V because again, I could be completely wrong. There could be a better way to do this, but, um, yeah. So, well, that was a really good answer and you really articulated like what I was trying to ask about. And I think, um, it's actually really easy to get access to a quantum computer, but it's not, it's really easy to get access to the interface to a quantum computer. And I think that what you, what you keyed in on is like, you need the morphology, you need the structure. Like, so in order to play with a quantum computer and program the quantum computer to do mortal computation, you actually have to play with the substrate itself and not just have access to the computer through some interface network. Right. Yep. That was key. Bingo. To kind of bring us towards the non-human cognitive modeling slash fourth paper, something coming up in that was all the different ways that we talk about interface and also the different ways we represent it. Like this is really creative and I've never seen the, um, blanket represented this way with a, with a zone and which connects nicely to the blanket index, but whether we're talking about the kind of strict analytical base graph Markov blanket or the quantum interface or just an engineering interface, input output, or flow diagram systems dynamics, or the kind of perhaps most general or accounting system of them all like applied category theory, we're describing these different facets of the interface. So that's one place to begin when modeling a non-human or a human cognitive architecture is its interfaces. It's like, if those aren't accounted for, we're either speculating information that we're measuring or observing or interacting with this cognitive thing, whether speculating about what we could do or about what we could measure, or we're missing things that we actually can intervene with and measure. So maybe how do we go about before even wondering about which formalism it's going to be described as, how do we think about the interface of a cognitive system? Well, I think that's, yeah. Well, you just brought up a really good point. I just want to point out that you said, uh, if we don't design, you know, the interface between let's say the internal states of the agent, which just to connect it to the cognitive modeling paper, you could think of the common model of cognition as the design of the internal states, right? It's the brain, right? And, uh, let's just keep it like the organizational principles of the brain. Because again, I think of CMC common model of cognition as a blueprint and, you know, cog engine, which is what that paper is about is a physical, well, it is a computational instantiation of some pieces of CMC. Um, and even that is still only a part of the puzzle. So, but you could think of that, that's going to be what's inside the Markov blanket. All right. Uh, and then of course, inside of it could be more Markov blankets. Cause we also talk about Markov blankets of Markov blankets of Markov blankets. And there's a recursive assembly kind of, but, uh, you brought up Daniel, if you don't design, uh, it could also be an engineered, you know, uh, permeable boundary with pumps and things that, you know, for example, remove toxins from the internals of the agent and brings in nutrients and energy, which we do talk about as well in the, in the mortal computation paper. Um, regardless of the formalism, you're all, if you don't account for it and you don't, so while cog engine does not account for it, it's still one of those like, well, we're going to just say, we're looking at some very, very specific internal states and we're going to probe it in a very specific way. Although we kind of have an agent cause we do use the gym mini grid. So there is a body, but it's like a really, really, really simplified body. Uh, you're missing out on a lot of intelligence and cognition that the boundary itself will give you. So even in cells, right? So this is to take what you said, Daniel, and you know, Jack, uh, boost it even higher. The boost, the signal, uh, is that I also believe a lot of the intelligent computations that we do can be, are offloaded, right? This is the embodiment thesis of cognitive science. The, the, even the permeable cell boundary, right? Of just single celled organisms or multicellular organisms. Uh, it does a lot of beautiful things that the internal states are not doing like, uh, consciously or directly with effort. May I should be careful with consciousness at different levels, but, um, the, that boundary, the Markov blanket offloads some of that compute that the brain or the internal states need to worry about. Um, and it can, they can sort of rely on what the boundary does. Now, of course, there's more to that story that even the Markov blanket doesn't necessarily need to address it. Uh, but there's the fact that there's also autopiesis. So the idea is that the system reproduces, it's self-reproducing. That's where, why there's a lot more, it's a paper that very much is like an iceberg. You can read the beginning and be like, that's a beautiful phrase. And then what's underneath it is this extremely deep. And I don't even think I, Carl and I have properly even captured all the facets that you could add to this. There's more like we just talked about the quantum interpretations as well. Um, so assuming that we have everything taken care of, that things self heal, self repair, uh, the, the boundary itself is in, you know, constant reconstruction, uh, you, you have this nice power or nice advantage that you not only account for how energy flows in and out of the system and things that you would normally not account for in our traditional agents that we designed today. Um, but you would also have the ability to offload cognition. There's even a domain. We mentioned it in the paper too, but, uh, you might've heard of morphological robotics. Um, and there's even classical ideas going back to even, I believe even Rodney Brooks mentioned aspects of this as well. And some of his early work, if you study the way that we move our joints and our arms, uh, you know, there's a lot of, uh, cognition offloaded into that act. And you can also get that in terms of robotics too. And just saying by understanding the, you know, kinematics that drive an arm, then now the robotic brain, if you will, the neuro robot only needs to say, I need to move my hand to this point. And then it can sort of trust its digital nervous system or rather the actual constructs and the robotic hardware to move the rest of the arm. Like he doesn't have to think consciously about moving the elbow, just needs to move the point. If you know from computer graphics, inverse kinematics, right? You only need to move the point of the hand to where you want to go. And then the rest takes care of the motion. And it looks, you know, it doesn't break the constraints of how like the bones are structured and the skeletal structure of the armature of the character. So, uh, I think you, you, there's even more to just model, not just, so we do talk about modeling too, because some of us, uh, or a lot of us might not have access to the hardware as well. Right. I think it's in the appendix. I think it had to be moved to the

supplement of the paper because it was already very long. Uh, yeah, it is in the supplement,
but we have this problem called the body niche problem. Uh, and besides the fact, it's a nice
sounding philosophical problem. There's this issue that, uh, robotic hardware or even organoid, you know,
constructs or biological substrates or chimeric won't be available to everyone. So if we want a
democratize things, we're going to probably need to model these digitally. And so we have this concept
called a digital morphology. So maybe we can take whatever principles of physics and other aspects
of thermodynamics. Maybe we can construct bodies that we can simulate for cheap so that we people
can work on mortal computers simulated. Right. Uh, although the ultimate would be you just like
blue brought up earlier about the best way to use them, a quantum computer would be to literally have
access to the more quantum computer and then design your internal states, your software that would run
run it, uh, literally on chip, right. Or on, on quantum computer, whatever the right word for that would be,
but they're entangled. So you're completely, uh, you're only paying the cost, the thermodynamic cost of
reading and writing to the memory right on the chips or right on the actual platform itself,
uh, rather than having these separations, right. Even including like operating systems and the
transfer of memory and a hierarchy. And so having the actual hardware would be the best. And I think
that's where, uh, you know, even the work kind of says this is where the future really lies, but
probably along the way we could probably work with digital morphologies as well. So long answer,
but just to say that your point is great and we could boost it even further by offloading cognition
to the body. And I also think, cause you said, what formalism would you choose? And I don't think
any one formalism is, you know, like superior per se, they give you different things. I think the
Markov blanket is nice because it connects to a Bayesian probability, and then you can kind of
at least characterize the relationships between states. Um, but from an engineering point of view,
I don't think you like directly have to design the exact Markov blanket because now you're dealing with
a graphical model and what would you do? What do we usually do? We usually choose discrete states and
the world's continuous. I've thought about this. Like I thought, okay, now that this paper has come out,
how do I build my own mortal computer, right? To show how we could do this. Uh, but you know,
I think Michael Levin has a great example, uh, with the Xenobots. Like that's one way to do it,
although it's very biological artificial. Um, but I think just designing boundaries
and sort of clearly explaining or accounting for the thermodynamic exchange and understanding like,
what are the resources that we have at hand, even in a very controlled environment might be the way
to design the Markov blanket, because then you get for free. If you have like, and I think we talked
last time about like a quadruped neuro robot, like spot or something and, you know, have it outside,
right? If you already choose your body and you put it in an environment, you've already now dealt with the
boundary and the niche. You just need to understand the properties of that, which is where we can leverage the,
the wonderful knowledge across domains like physics and biology and chemistry,
understand the properties of that body, right? Understand the properties of that niche.
And then all we have to work on is the internal state design. And how does that interact with

maintaining persistence, right? For example, battery level, power level, that's a homeostatic variable or a metabolic variable. There you go. Now you can build your agent about keeping that variable, uh, in within a certain range, right? Because homeostasis is about a set point.
So may have a target or actually the, we changed the word to something more powerful, hormiohesis, which is homeostasis over time, right? Because set points might change like in flux to the environment. Um, so that's, that's a kind of a cool concept as well. There's like a word for everything I learned when writing this paper. So many concepts already exist. All we have to do is just stitch them together. And this is one way to stitch them together into a nice little framework. Um, so did that make sense?
Yeah. I'll, I'll make a note and then blue, I'll be curious to see what you think about this. I think with what you've just described, you've, you've almost given some of the most specific ways to think about what is offloadable. We've talked about extended cognition and about having passive artifacts in the niche like paper and pen, and then having active artifacts in the niche like autonomous agents, or whether they're material or, or just digitalized. So it's interesting that by putting certain computation on the boundary or on the interface broadly conceived with both, uh, material thermodynamic and a informational semantic component, then it's like, yeah, the nucleus doesn't decide about the cell morphology. And so it's like, how many layers can you put in between in between so that the computation goes out to the interfaces everywhere's an interface from somewhere. And that's where the work is really done. And then there's just so many other great points, but what do you think about this kind of conclusion to 2023 live streams after long arcs with thinking about embodied cognition and cognitive architectures?
Um, so, wow, what is offloadable? Uh, I have a lot to say about that. Um, uh, so like in terms of what the body does, um, and let me know if my internet cuts out and I'll shut my video off. Um, but in terms of what the body does, like, it's like, you think about things that are automatic, right? Like the heartbeat and the breath, um, and stuff that like we just ignore. And like, I mean, if you really, you could do biofeedback and slow your heart heart rate and like breathing, you can definitely control.
I mean, it's very obvious that you can control it. Um, but, but like, I wonder about these things, like what is offloaded to the body that we're ignoring, right? Like if you offload too much, you know, like we have like gut reactions to things like, like, you know, these like sensations, and there's like a lot of computation happening within our own morphological structure that maybe we're not conscious of. Like, so a lot is, is unconscious, but then like, are we overlooking stuff that we should then be paying attention to? Like, like, I wonder what is the price of, of offloading?
Hmm. I mean, that's a great question. Um, I mean, I don't know. To be fair, I can't say I know the exact sweet spot, because that would also mean that I have, uh, the exact instantiation of the mortal computer that I wish to build, right? Um, you know, the answer is we need to study that.
Uh, that's the short answer, the slightly longer answer. How do I say this without letting myself ramble?
Well, so maybe there is a point where we cross, where we have, let's say humans, uh, animals, even any, any entity has failed to find the sweet spot. Like you just said, Blue, uh, even we humans,

humans, when we do something, you know, we have a gut reaction and we might tend to listen to that gut reaction rather than doing some other type of processing, like thinking critically or logically and counteracting it. We also have emotions, which I think is very important, which we don't really quite have that, uh, in today's AI, but, uh, we do kind of have a little bit of some good work has been done in cognitive science and cognitive architectures, but I think that we might have already crossed the sweet spot because some of us listen more to our guts and our emotions than logical, logic or reasoning, or, you know, planning things out more carefully or listing pros and cons. Some of us might not listen enough to our gut feelings and we listen too much. So I think the diversity, uh, in even within any entity such as a human is vast. And so maybe someone has the optimal sweet spot. I don't know who does it varies, but it doesn't matter. So it's kind of interesting. We it's, it's, it's an interesting question to quantify and worth looking at, but at the same time in designing a mortal computer, let's say I failed to design the one that has the right balance. Let's say I designed a computer that mortal computer that leaned more heavily towards gut reactions, uh, than not. Well, I would still be thrilled that we built such a computer because while it would be great to find and characterize these set points or sorry, these boundaries and regions, um, we would still have the premise that look, life is built on these hierarchical or rather layers of complexity. So homeostasis, homeo, sorry, homeorhesis and homeostasis, which sit at sort of the bottom of what we talk about in the paper. Um, and then underneath that are like, you gotta be careful a little bit, but they're like metabolic pathways or what we, or Carl and I call essential primitives or variables, which are kind of like the materials and the, the very base constructs that maybe we, you know, manipulate to survive with. But above that are things like allostasis and complex processing. And of course we put at the top of our hierarchy, uh, auto pieces, but technically, uh, the idea is that everything would be self-repairing and healing. But if we put the growth and dynamic change aside for just a minute, our allostasis can be argued, and it has been argued to be all kinds of things. It can be, uh, oh, we have little neural circuits that try to, uh, predict when homeostatic processes are going to need resources. And so this is where you can see it in animals and, uh, insects and other types of, uh, creatures that, oh, I know that, uh, the seasons are going to change. Like birds, they fly to another, and they, so they have these kinds of instincts built in, or they have some sensors in circuit processing and say, I should probably go somewhere else where I'm not going to die for the winter. And so then they flock away, right. Or they fly away. So there's already some higher level behavior built on there. The reason I bring up that, just to tie it back to that balance point that you bring blue is that we would have the opportunity to design those neural processes that could override, for example, your gut reaction or your more primitive instincts. And so this is what's kind of nice about the mortal computer is we would want that ability to do planning, to do higher level thinking. And this is again, what connects nicely to the cognitive architecture, right? If we have a more developed, let's say brain region area, right? We have different types of memory, like procedural declarative memory. Um, now we're able to start to think at a higher level. That's not just purely driven by our instincts. And so the 5d cognitive theory, uh, in the paper, elementary cognition, I don't think is the only thing you need. That would be like, I only rely on my gut instincts. I am primitive in the sense primal. I only want to live and I will do whatever it takes to live. But for example, if you live in a society of mortal computers, which is one way you could think of humans, I'm not going to start calling my friends mortal computers, but we live in a, in a relationship. We have a, we are a complex system in ourselves. Well, now we're not just only thinking about, uh, exactly like I, I don't want to die right now. Right. Cause if that's how we live, then, you know, we go back to, uh, kill or be killed law of the jungle, right? We are just sort of fighting amongst each other. So I think while it's hard to answer what's the sweet spot or the balance point, I would still think having a mortal computer that even has the ability to engender those primal gut instincts would be very powerful. And then we could work on thinking, well, what are the right neural processes to counteract that? Right. To add the critical logical thinking, what constructs would have evolution evolved to allow us to counteract that? Um, does that make sense? Cause I think like, it's almost like it's a good question to ask, but I don't think it's, uh, it's more like it would be, it would be profound if we had something that leaned a little too heavily on its gut instinct. And we say, okay, but now we have these allostatic processes. We're just missing more of that higher level behavior to counteract that, uh, because we humans are able to, for example, ignore even sometimes our primal needs, our desire, not maybe completely desire to persist because some of that's very deep, low level. Um, but like we even have the act of like heroism, right? You know, like I'm going to go save someone that I care about and that will end your persistence. You will thermodynamically disintegrate when, you know, I go into the fire. Right. So we have these higher level cognitive processes. We're going to need to design those two or let some evolutionary process, let those emerge. So, um, I would still, we just don't have the low level either. And we, we might not have the full picture of the higher level, but right now, a lot of artificial intelligence doesn't even have that, that primal persistence. Right. So I think I would be thrilled to have that problem and say, oh, we might make this real real real real real real real real real real real real real real real real

real real real real real real real real real real real real real real real real real real real real real real
he talks about the entanglement of thought and emotion,
which I think like thinking about quantum computing and quantum entanglement,
like the entanglement between thought and emotion is just like a really
interesting concept to me,
or just an interesting way to think about like who wins or like,
can you have one without the other or one always knows what the other one is
doing or yeah.
And the problem we have in,
I agree we have that entanglement and it's fascinating the problem in
artificial systems or technological artifacts, which by the way,
another theme of this paper,

and we sort of wrap it up is it's like a path to,
it's going to sound a little odd,
artificial sentience.
And then that also opens up the whole moral issues and ethical dilemmas.
But, but to even get there,
I would argue we're missing the half of your entanglement, right?
We're missing more the emotional side,
which I think also stems from,
you know,
even the lower level processes that kind of give rise to the emotional states.
And, you know,
there's,
there's a lot of physiology and a lot of physiological connections between like
the chemicals and things in our body that also help drive those emotions.
And I think we need them.
And I think artificial,
technological artifacts are missing at least.
I'm not going to say if we've tackled sufficiently the higher level cognitive
behavior,
because,
you know,
there's still the work being done in cognitive architectures and cognitive modeling.
Although I also think that we're,
because we're missing half,
which is the emotional side of things and the biophysical drives and manipulations,
we're missing that entanglement.
So the design of our intelligence system is sort of like detached.
We're still in many ways in AI.
And we also mentioned this in that paper.
You might've heard of brain in a vat,
right?
Which is the idea of what you can think of intuitively.
It's an old cognitive,
it no one really ever said it.
It's more of like a criticism of old cognitive science,
which is to say that neural processing,
our mental representations,
our decisions and that we make and our plans that we make and our complex logical order processing

are completely divorced of the body,
right?
It's the extreme.
It's essentially mortal,
immortal computation,
which is kind of another fun thing we pointed out.
But in the cognitive sciences and philosophy of mind realm,
and because we've been designing things from that perspective for so long,
we're missing that entanglement.
And I think there's some of the roadblocks we will face in the future,
some of the things we can't necessarily build,
maybe the holy grails that still remain,
like,
you know,
higher order logical reasoning and complex planning that,
you know,
transformers and the big neural models can't really quite do today,
despite claims that are being made from the public relations of different companies.
I think that we're missing that other piece of the equation that could have helped inform and make those processes that we design better,
but we're also missing that relationship.
So that entanglement blue is inherently not there,
which is there in,
for example,
human entities and animals and,
or in living beings,
right?
We have that interplay,
and then we have the ability to decide if we want to override it,
but technological artifacts don't have that.
There's nothing to override,
right?
So,
um,
at least I'm going to make that claim.
I'm sure that there's probably,
there is this world called effective computing.
Um,

but I'm not a hundred percent clear if whether or not that has all the ingredients that I would say need to be in place to build that 50% of the entanglement of emotion to higher level cognition and reasoning and non-emotional reasoning.

Um,

so,

and we need that.

So I would love to have that.

And then we could then quantify that entanglement,

right?

And talk about the relationship or proportion of emotional processing,

or even low level biological processing to higher level or cognitive processing and planning and decision making.

I don't know if blue heard that,

but I see her thing green.

So,

okay.

Yeah.

Yeah.

Wow.

I brought up an image.

This is a figure 1.2 from the textbook with a high road and the low road.

So,

the low road,

how it's constructed and that's Bayesian,

but we could consider other ways that things are actually built like JF Cloud,

you're working on the symbolic active inference.

So programs from the bottom up,

or there could be different heterogeneous elements.

We talked about quantum and neuromorphic.

So,

we talked about quantum and neuromorphic.

So,

as many material substrates as there are material substrates to be inclusive of what ecosystems actually consist of.

And then first principles thinking.

First principles approaches,

including the free energy principle or unified through consideration with the free energy principle.

all of these terms that include complex systems, phenomena,

self-organization,

autopoiesis,

like you mentioned,

and how the current narrow or whatever it is,

computer science that we continue to ride inertia and acceleration on from the kind of von Neumann architecture and scaling in materials with a lot of costs.

Like that is almost missing these two complements.

One of them is the top-down organizing principle,

though there are organizational principles within computer science.

Of course,

it's a formal field,

but unified principle.

And then almost like in a,

in a very interesting relation that,

that I hope people can explore later is this other complimentary relationship between,

I mean,

to put it coarsely,

like the heart and the brain and the integration with the hands and about pragmatic value and the affective,

value.

And blue,

you asked like,

what if we're,

what if we're like least aware of what we offloaded?

It's like,

that's,

we're,

we're only aware of what we're not offloaded.

So every,

every,

if we just take a,

a mental survey of what we know,

that's the remainder of what we don't know.

And that's a very different,

very integrated,

compressible perspective that then can unroll a lot of different ways.

That isn't just like,

we have to do what we're doing now,

but just more.
Well,
and just a quick point.
There's also this beautiful concept called,
and it's being embraced in cognitive science more recently,
because it was even the theme of COG-Sci,
I believe last year,
cognitive diversity,
you know?
And so,
and that's the beautiful part of embodiment,
the embodiment thesis in cognitive science,
which is,
it can give you an explanation of all the different ways in which we might
have something more offloaded than,
you know,
not into the different varying degrees.
And so even maybe the question sometimes isn't what's optimal,
but rather,
can we capture the wide array of different amounts of cognitive offloading?
Because if you change the body,
you change the cognition.
And that's like the embodiment thesis in its natural,
and this paper echoes that extreme kind of view a bit,
that if you change the body,
you're going to change the thinking and fundamentally,
and you,
we know this even,
even internally within the same body,
if you are feeling sick,
if you are feeling hungry,
or you have your,
even if you,
you can fight these things with a top-down approach to some degree,
but it will affect,
for example,
your focus.
If you're sick,

it's harder sometimes to write a paper.
You have something called brain fog,
right?
And so that's where the body is so intimately involved in our biophysics and our physiology,
so intimately involved in our everyday processing that,
and different people are affected to different degrees.
I mean,
I'm sure there's someone out there who's like,
no,
I'm maybe they had,
for example,
a really bad cold.
And they're like,
no,
I was able to get the same amount of work done as always.
Right.
I mean,
and that would also show you that it's going to be vastly different,
even within the same type of entity.
And so that,
that's something cognitive diversity,
I think gets captured by this type of thinking.
And it could be really interesting,
you know,
from using the premise of a mortal computer to sort of construct systems that are like that.
And by the way,
the mortal computer rests on that dice diagram that you have there,
you know,
with the active inference and the free energy,
it sort of tries to hit most of those main points and sort of like,
agglomerate them into,
well,
we even propose a definition,
right?
I mean,
is it a complete definition?
I'm sure there's extensions.
There's probably corollaries to add to the mortal computation thesis.

But I just wanted to throw that point in there because I think that that's where it's that,
that how much we offload will also change,
even if it's like,
maybe not optimal,
but you know,
that would be more dictated by an evolutionary process.
Um,
but you have different entities and different organisms that offload to different degrees.
And depending on your body and what affordances it gives you,
what,
what abilities it offers you to interact with affordances in the environment and tools and things like that.
Um,
if I had eight legs,
I'm going to be able to do drastically different things than,
you know,
with my bipedal body right now.
And morphological robotics usually explores some of that.
Um,
what can different bodies,
beyond the humanoid style,
uh,
provide in,
for example,
quadruped robots,
which they take,
you know,
another thing in nature,
like,
oh,
let's say animals like dogs and cats,
but they're wonderful for like search and rescue,
right?
Whereas a bipedal system might struggle to get into certain spots.
But if we had a quad,
and there we've changed the body and hence we've changed the cognition and the mental neural processing.
Yeah.
Yeah.
A lot,

a lot,
just even from the outside of robotics,
I hear about cognitive robotics.
So the adjective is saying,
well,
we're pulling back to consider cognitive architectures.
And then there's another layer,
which is like,
well,
how does it get there?
What's the developmental trajectory?
And then what is the ecology of similar or different minds and,
and bodies?
And then what's the evolutionary process?
You can keep the,
like you,
you,
you,
again,
in,
in,
in kind of distinction,
describe the immortal.
It's designed as if immortal,
but of course,
in reality,
the Linux operating system or any given language model is mortal.
You just unplug it because it's energy resources aren't accounted for.
So it's kept in this suspended bare life state where it's at the behest of something,
else.
And so then that like unsustainability propagates through a system causing externalities.
So ensuring that the externalities can be understood where they are,
it leads.
So we expect to scalable,
interpretable systems that don't just hide the semantic ambiguity and the material
components.
They're so abstractly considered.
And it's so unintegratable with emotive components that it feels like a dead end.

So,
I hate to keep talking about mortal computation,
but we're doing it.
Well,
it's your,
it's your guys podcast.
I'm happy to lab about anything.
I just want to make sure that if you wanted to talk about those papers,
by the way,
we did talk about those papers last time too,
and they're integrated,
right?
Maybe we can relate some of them.
Like I keep bringing the cog engine paper,
not because it of itself is mortal,
but I argue that it is a piece of that puzzle.
Like if I were to go,
yeah,
that architecture there,
that's not a mortal computer.
However,
it is a cognitive architecture.
And I think it plays a role in the design of systems,
right?
For example,
maybe blue,
it helps.
It would help us,
for example,
to design higher level processing to counteract some of the lower level emotive,
you know,
decisions that an agent would want to make.
So I just want to make a mention.
I'm fine with talking about whatever.
I'm the one that shamelessly plugged it.
I just actually wanted to plug it.
And then,
you know,

we could go from there.
If you,
but they're,
they're,
they're intertwined.
Right.
It's not like my thinking is,
Oh,
when I work on cog engine,
I turn off the mortal computer.
It's like,
no,
they're all,
they're all swirling around.
It's all part of the same thought.
So if you want to keep talking about it,
be my guest,
I'm your guest.
So I just,
I just have a question about like,
when we talk about scalability and interpretability,
I wonder how much interpretability is due to the fact that,
you know,
we're missing that emotive.
Setup.
Right.
Like,
so if we have some kind of interplay between like persistence,
emotion,
thought,
like if there's some kind of,
you know,
struggle,
like who's winning the war when we make a decision,
when we choose to act,
are we going to lose interpretability when we just start to design mortal computation?
Like,
or is there a way that we can keep interpretability?

I mean,
I guess we do all the programming.
So,
you know,
you keep there,
is,
is it going to end up being more black box?
If we have like,
I mean,
a human is way,
like,
I don't know what's good.
Like I just compute,
you know,
I don't know what's going on.
I mean,
I can try to check in and like,
where's this coming from?
Is this my logical brain?
Or is this from my heart?
Or is it just like,
you know,
a motor like kinesthetic memory or that's driving me to act in a certain way?
Like I can try to dissect it,
but even it's hard to know my own like computational process.
So when we're looking at like designing an artificial system,
are we going to lose that interpretability in,
in the design?
Is that like built in?
Well,
fascinating point or fascinating premise.
I think it depends not to bust up,
the famous,
it depends answer.
And any of my students listening will be like,
well,
there he goes under the shield of it depends,
but it really does.

And I think it depends on what interpretability you're looking for.
And I'm also going to mix,
and it is different to some degree,
explainability.
And I'm going to bring in that word here because there,
there's those that work in like interpretable AI versus explainable AI.
And they can get pretty mad if you like say,
oh,
it's the same thing.
And it's not quite.
So in designing,
let's say a mortal computer,
let's even say like a con.
So in the paper,
we also have sort of like the beginnings,
because by the way,
we are encouraging everyone to contribute.
We have sort of like a low level or what we call a homeoheetic or homeostatic
mortal computer.
And then things that have other processes are allostatic.
And things that are at the top level,
the complex computer would be autopetic.
And that would be like the ideal.
That would be like,
oh,
well,
there we have a human,
right?
We have an animal.
We have at least a sentient organoid.
Yeah.
If you're looking for Daniel,
it's like after it's in section five,
under the definition.
But if you wanted to see that little,
it's like it's sort of a taxonomy.
And then we use it to evaluate modern day mortal computers.
No,

scroll.
It's not a picture.
No,
no,
no.
It's under the definition.
It's,
it's a bullet point.
Scroll down.
You were close.
It's weird.
All right.
Yeah.
Go down.
That's the definition.
And then you scroll down right there.
So there,
yeah,
we,
you had it,
you had it right there.
Scroll up a little more.
No,
a little more.
Scroll a little more.
Homeostatic,
allostatic,
autopedic.
So we sort of give a little bit of a characterization of like mortal computers of varying complexity.
So I'm referring to this just in answering Blue's question.
If I designed the ideal MC,
mortal computer,
autopedic,
right?
So it has all the ingredients we want.
And then you're saying,
is it interpretable?
Do we lose some interpreting?

I would argue not completely.

You will from a,

so the,

the parts of interpretability you preserve,

and I think it's even better in some ways than your run of the mill transformers and large neural models that are in operation today,

is when,

is when,

for example,

chat GPT or GPT-5 or stable diffusion does something really interesting,

like,

oh,

it can do this downstream task that I never expected to do,

like zero shot learning,

which,

you know,

arguably is very interesting and promising that these models can do some elements of this type of adaptation.

We can't answer why that happens.

The best answer we have is,

well,

it's because we keep throwing more and more parameters at these models,

more neurons,

more synapses that connect them.

And so if I toss more Legos at this,

it'll just emerge,

you know,

through emergence,

create complex behavior,

and we get these nice little properties.

And it's interesting because that does relate to complex system,

complex adaptive systems theory,

like John Holland's work in the 90s and the Santa Fe Institute,

and other people that have worked on dynamical systems and complexity theory and how you get complex structures that emerge from these very simple structures.

And self-organization was actually on Daniel's hierarchy.

It was like a one piece of an organizing principle.

So that part is like the part where it might be hard to completely interpret.

Why does this behavior emerge?

Because it is this complex,
multi-level interaction of parts.
And you could at least say it comes from like,
and we do talk about it too in the paper,
because we do talk about self-organization.
You can't without,
if you're going to talk about any living system,
you have to go into self-organization.
Upward causation and downward causation.
So upward causation is like,
the little parts give rise to complex emergent parts,
and then their relationships to each other.
And then the top-down causation is,
you have these higher level parts that have now laws and physical relationships
and implicit relationships with each other that exert like a modulation,
modulatory effect on the lower level parts.
And so that's why you have top-down, bottom-up,
like causation,
arrows of causation and interaction and constraint.
So to give rise to a complex system.
And so you can point to those pieces and say,
well, that's the reason why certain behaviors probably emerged.
But then you get into non-linear systems,
and it's very, very hard to even, you know,
mathematically analyze them,
let alone, you know,
come up with intuitive qualitative pictures of what's going on.
But the part that you also get in designing a mortal computer
and like the cog engine architecture that Daniel has here on the slide
is you get interpretability in terms,
at least the structure and some roles,
especially if we're talking about like the internal states.
Like the reason I like the common model of cognition,
and there's other, you know,
theories and blueprints out there.
It's not to say it's the only one.
But the reason I really like it
is it sort of cuts up or cleaves the mind

up into these general blobs of responsibilities
and has some relationships.
It doesn't commit you to any implementation,
as we talked about at the podcast point one,
but it sort of tells you like,
oh, procedural memory,
it's going to be doing this very specific thing, right?
It's going to learn how to memorize sequences of actions,
kind of like a reactive, physiological, physical, reactive components.
And, you know, the way I interpret it
was the transition dynamics model of active inference,
or let me build a predictive coding little structure
that tries to guess latent states.
But then you have like declarative memory,
and it stores the facts of the world,
things that, you know,
I bind the name of a person to their face,
and, you know, I'm storing these little bits of knowledge
into my representations and declarative memory.
So you get the interpretability,
at least at the level of,
I know what these modules are doing.
I probably could investigate and probe,
in even a computational fashion,
the contents of memory,
cognitive architectures in general.
And so that's the part that I like.
They bring at least a level of interpretability
because you have this organizational principles,
you have responsibilities for different modules,
and even classical cognitive architectures
have things like production systems,
and you can encode knowledge into them.
There's good work on like, you know,
if-then statements,
and you can kind of break up knowledge
into at least bits and pieces of logic,
and we can very easily inspect that, right?

And so the,
while that might not be computational
in the most efficient way,
we could probably do mappings
between the neural representations,
since they're essentially mimicking
what like ACDR and SOAR
and very classic prominent cognitive architectures
are trying to do to these production systems.
There's been good work,
even from Yahshua's group.
He had a paper,
neural production systems,
which I kind of like that idea
about how do we emulate that piece of cognitive architectures
with, you know, just neural networks.
And so we can take advantage of that mapping,
and we could probably get some decoding going on
and interpret some pieces of our agent system,
mortal computer.
And also, by the way,
the very design of the mortal computer,
you actually would have a clear picture of like,
well, these are the essential variables
or the metabolic pathways.
Here are the homeostatic processes.
These are the things I defined
that define the life of this system,
or otherwise it will die, right?
And then the allostatic processes live above that.
And so that's what that like cool picture
that I really like in the middle of the paper,
the biophysics,
shows you like the hierarchy
of different types of processing
that would happen from biology.
And then the cybernetics tells you,
what is it going to do

from like an analysis point of view?
It's trying to find stable states.
It wants to be ultra stable,
to use Ashby's and the cyberneticist's old terms, right?
And that's, you know,
I'm able to very quickly satisfy my constraints
so I can keep living and not, you know, die.
And essentially preserve
or have enough internal variety
to counterbalance my external variety.
And that's the cybernetician's
kind of framing of what's going on
in organizing systems.
And so you get that interpretability
in terms of the design.
You know the parts,
the mechanisms behind it,
but you will lose the piece of,
like, why did that behavior
specifically come about?
It's going to be a nonlinear process.
So you will lose it.
And I don't think you can get all of that back.
Here's the last point.
I said at the very beginning, explainability.
Now this is different than interpretability.
And the way I,
it's going to be my digested version
of explainability.
So Blue, you were saying to me earlier,
I might be able to inspect
the contents of my mind
and come up with a reasoning
for why I did something.
And then you mentioned that,
well, it could be wrong
or, you know,
I might not completely capture it.

And I agree because we do this
in all the fields, right?
Like, especially anything about the mind.
I try to think about the contents
of why I did something
or what was going on in my brain
or break down my process
and then try to turn that
into a computational model
that then explains like the actual entity.
And it might be wrong.
It might be biased.
However, there's something cool
about what you doing that.
That's the explainable part.
You Blue might not know the true reason,
the real reason of the universe
or the biological reasoning
or the thermodynamic cognitive reasoning
of why you did something specifically.
However, you can provide an explanation
or a rationalization of why
why you did it.
And while it might not be perfect
and we shouldn't, you know,
take the account as wholesale
the exact truth,
you are able to communicate
the basis of your decision.
So you are able to use your construct,
your biology,
your neural processing
to formulate.
And then, of course,
we also have this thing
called communication.
You use the other parts of your body
to transmit those sound waves

to me and Daniel
across a tool, Zoom.
And we hear what you're saying
and you say,
oh, when I wanted to solve this problem,
I did X, Y, and Z.
And what we would find is then
if we did X, Y, and Z,
maybe that's not exactly
the true explanation why you did it,
but I would be able
to solve the problem now.
And we teach, right?
We pass on this knowledge.
And so the explainable part
might arise.
I think that might counterbalance
the lack of interpretability.
So while we might not be able
to say exactly 100%,
I still argue we have
more interpretability
than monolith structures
like transformers today
where we just say,
eh, it emerged.
And that's cool, right?
I mean, there is probably analysis.
I don't want to like
completely dismiss transformers.
There probably is interesting work
on maybe identifying
what parts of the network.
But from the design principles
or the inductive bias
that evolution gave us,
I think a cognitive architecture
gives you some interpretability

in a structural way.
But then you're going to get
a mortal computer
that would probably be able
to explain to you
why did it do this?
Because why?
The 5e cognition theory
that Carl and I have
doesn't, it absorbs
or expands 4e cognitive theory,
which includes embedded.
And the embedded cognition
is really important
because that's part of that
like other diagram I had
where I had the slices
of a pie of cognition
in a mortal computer.
And among there,
it's saying,
I have to interact with other,
well, I had other neural robots,
but I'm talking to you, Blue.
And that means that
I have to be able
to communicate
some approximation
of why I did something.
And there is an explainable,
you know, there's explainability.
And I communicate
this reasoning to you,
you give me feedback,
and, you know,
we kind of come up with
a solution to the problem
or a better understanding

of something.
And I think that's where
the mortal computer
in its full glory
would give you, right?
Because it would be able
to tell you why.
It would interact with you
because it also needs to.
Because you're part of its niche.
You affect its survival.
So it can't sort of like,
oh, okay,
I'm just going to ignore you
because actually
it might be really important.
You might be important
for it to achieve its goals,
which could be
to allow it to get more power
if we go back
to the primal connection.
There's a lot of jumping around
going on here.
But you are an essential part.
We, humans,
all of us,
are an essential part
of what a true
mortal computer would be.
This is what makes it
artificially sentient too, right?
And it has to live
in its niche
and the niche,
you know,
can get very complex.
It's not going to be

a laboratory
that I put it
in a little white-walled room
and, you know,
oh, okay,
I've emulated life.
No, ultimately,
you would want this
to integrate
with human society
and to interact with us
because there's things
that you get
that with a group,
with a collective,
with social norms,
that you don't get
living in pure isolation.
And so you're going
to get that.
It's not interpretability,
but it's explainability.
And I think that
that communication
and that part
might counterbalance
the fact that
we're dealing with,
as you rightly point out,
Blue,
an extremely nonlinear,
complex system
that it's going
to be very hard
to say exactly
why did it do
what it did
based on inspection

of its contents,
even if we designed it.
So the explainable part,
I think,
will counterbalance
that to some degree.
Wow.
Sounds like psychotherapy.
Psychotherapy
for the mortal computer.
There you go.
Yes, we need it.
There's so many pieces
to add in.
I barely know
where to go.
Yeah.
Yeah.
It's a lot you brought out.
I think the connection
to complexity science
is very good
because that's where
we can ask
and have a laboratory
for like,
well,
what's the account
you can give
of this double pendulum?
What's the account
you can give
of Wolfram's Rule 30?
And by that,
we can contextualize,
yeah,
how many experiments
of what type

would we expect
before we could understand
a fruit flies decision
in a T-Maze?
What about a human
in this or that?
And then this tension,
but fundamental,
so we can quickly
come to adapt to it,
that there's things
we do and don't know about.
It's one of the quotes
beginning a chapter
in the 2022 textbook.
I think it's like
a Minsky quote.
We're aware of least
what our brains do best.
And so that whole
like searching
under the street light
with the light
of attention
that every agent
is only engaged in
as a fundamental constraint.
That's one constraint
is the attentional,
informational.
Another constraint
is the material,
physical niche,
which is always actual.
And that's,
that's connected
to the environment
totally fundamentally.

Like there can't be
a discussion of what
ant colony foraging
strategy works best.
Even for one field location,
there's going to be
different ant species
with different
total strategies
and different nest mates
with sub strategies
as modeled by.
So you've embarked
on a wildly
fundamental task
that combines
a lot of the
discussions happening
in active inference
with the broader
cognitive sciences,
broader complex systems.
and it's definitely
interesting and relevant.
So I really appreciate
that you share it this way.
I'm happy you guys
took to it so well.
And I certainly want
that idea out there.
So now,
now that there's a,
what do you call it?
A proper treatment,
an initial treatment
of moral computation,
please spread the word.
You know,

I want,
I want to hear other people
come back to me like,
oh yeah,
I heard about
the mortal computer
and I'd love to see
you guys
and everyone else
try to take a whack
at designing
even the most
primitive versions of it.
Because I think
this is like,
I don't know
if it's quite that,
but it almost feels like,
uh,
like a little baby step.
This paper is,
uh,
is,
uh,
intensive as it has been,
uh,
and informative
and so fun
working with Carl on it.
Um,
it is just a step
in a big direction,
as you just said,
Daniel,
and it's a wild,
wild place.
And it also goes
at odds with a lot of,

like,
maybe this group here,
you know,
we're a lot more,
uh,
what do you say,
receptive
and excited
by these prospects,
but it does go
even to challenge
a lot of
what we hold dear
and near and dear
to ourselves,
which is the,
we really do like
our immortal computers.
And,
you know,
there,
there's,
there's gonna be
trade-offs to that.
And the paper does
talk about that.
It even starts by
trying to address
the reader
by saying,
the reader might ask,
you know,
themselves,
you know,
why do I really want this?
Because it's also
very brittle.
I'm gonna just be

clear too.
But then again,
humans and animals
and organisms
are also brittle as well.
Um,
but why do I want
this?
Right.
And,
you know,
and then we had,
and we motivate this
with like,
uh,
three kind of stabs,
which is thermodynamic,
which is the most important.
And by the way,
the free energy principle,
um,
and physicists in the room
might say,
well,
wait a minute,
free energy,
that's thermodynamic.
Why?
This is not quite the same thing.
There's the mixture
of information theory
in here as well.
But,
you know,
the thermodynamic free energy
is a really important
piece to this puzzle.
And the,

the way that computers
are organized
is sort of what
this challenge is.
And,
you know,
it also,
we have tools
to attack this,
like neuromorphic chips.
Wonderful,
wonderful platform,
uh,
for,
uh,
you know,
a mortal computer.
Uh,
and we do talk about it
in the supplement.
It got moved to the supplement
because,
again,
it was already
a little bit too long.
Um,
so I talk about,
uh,
neuromorphic architectures
and the,
also neuromimetic algorithms,
which kind of connect
to the papers too,
like predictive coding.
It isn't like front
and center.
I don't,
we don't try to like

throw it in your face,
uh,
even though it is a,
uh,
one way of thinking,
uh,
a consequence
of free energy minimization.
But,
um,
and so we talk about
these things,
but I think
it challenged,
we also talk,
it also challenges
and gets us to think
about existentialism,
uh,
and life and philosophy
of life and death
because,
uh,
we,
there was even citations
to Heidegger,
uh,
and,
uh,
and all kinds,
and Sorin,
and all,
all kinds of wonderful thinkers
too that have just thought
about philosophy of mind
and how,
uh,
our identities

and our,
our thought processes
are just naturally
influenced by the fact
that we know,
uh,
that,
you know,
things are going
to possibly end,
um,
and wonderful thinkers
and there's also been
this concept of like,
and again,
and this is a little
Heideggerian,
but,
uh,
you know,
your,
your being,
there's,
you know,
this famous book
being in time
and you think about,
well,
you know,
at the end of time
you,
you have to understand
when you live,
you're projecting
your being to there
and then going backwards
and so there's a quote
that kind of starts

the paper from
Sorin Kaikago
about,
uh,
you know,
we,
we look at life
backwards,
but we must live it forwards
and there's that interaction
with knowing
that this,
this rodeo is,
this play
is not going to last forever,
um,
which is a motivator
for why we do a lot of things,
including teaching
and passing on,
uh,
to the next generation.
And the other part
was again,
cognitive science.
I think cognitive science
got a little bit
of a spotlight,
uh,
because it needed to.
I think the embodiment thesis
inactive cognition
and then Kirchhoff's
and,
and,
and,
and his,
and his collaborators

life,
mind,
continuity thesis
needed to sort of
be foregrounded
a little bit,
um,
because I think that's
another piece of this puzzle.
Uh,
and it kind of ties
everything together
about how to think
about minds and brains,
but it all challenges
what things we're,
we're very used to.
We're very used to
when we design,
especially artificial intelligence.
Okay.
I,
I trust this hardware is good.
I'm going to try
to build some software
and I believe that I can
design all my processes,
uh,
from,
you know,
from scratch
without even accounting
for maybe what,
you know,
evolution might've given us,
uh,
we are seeing some of that,
especially Bayesian,

uh,
framings have some of that
in there,
right?
The idea of a prior,
uh,
I think is important,
but a lot of modern day
neural systems
don't 100%,
I think,
foreground that,
uh,
and play how important
these priors are
to sort of like
driving your behavior.
And so that's,
what's going to be interesting
is that this is truly
a weird place to be.
And it also really says
you gotta have a body.
You gotta have an environment.
So you're not done.
If you say,
I have this wonderful algorithm
that I think emulates
some piece of intelligence,
that's wonderful,
but that's not going to be
the end of the story.
In fact,
you're far from the end
of the story.
You need to be integrating
that into a brittle body
where that process

that you've designed
is also going to be shaped
now by the substrate,
right?
And it's interactions,
it's thermodynamic exchange
with the environment.
and the environment itself
is going to change
how it works.
And so you sort of get
this malleable,
flexible kind of framing
of AI as well
that intelligence is,
you just no longer
can divorce it
from the mechanics
of interaction
with interacting
with the world.
You cannot divorce it
from the thermodynamic exchange.
That's how we're going
to get energy efficient AI.
I think that's like
the other main message,
but it's challenging,
right?
And others will say,
well,
I have quantum computing
and it is very promising.
It's not any dismissal of it.
I'm just going to do
the same AI
I've always done,
but faster.

And do I really need
to look at,
you know,
how life emerges?
Do I really need to look
at how neurons
really do things?
You know,
it'll be interesting.
If nothing else,
it might be a little bit
of a race going forward.
Maybe it'll be
the mortal computing
still,
you know,
on its pedestal
being the dominant paradigm
and mortal computing
being rather niche
for a while.
I mean,
it's kind of the same thing
with the free energy principle
and active inference.
I like to think of
the mortal computation
as,
you know,
another,
a big expansion
of free energy
and active inference
because,
you know,
active inference
is a corollary
of the free energy principle

and at its,
mortal computer's heart
is the free energy principle,
right?
It talks about
the non-equilibrium
steady state,
the mess
and that's kind of
like at the core
and we cast,
there's the other parts
of the paper,
there's a lot of parts
of the paper
about inference,
learning,
and structure
and you cannot ignore
any of those pieces
if you're going
to build a proper
and that's actually
part of our definition.
There's an entanglement
there.
That's a whole
another conversation
but,
you know,
it's like an expansion
of the free energy principle.
It says,
you know,
what would,
where does this go next?
And it's really like
how free energy

often is used
in the literature
to explain
other phenomenon
and like,
for example,
Carl and many others
try to use that
to do things
in neuroscience
and neurophysiology
and computational psychiatry.
This is what would happen
if you take
the free energy principle
and you merge it
with hardware considerations
and morphology
and place it
in the world of AI
and,
you know,
say,
what if our goal
is for in silico
artificial sentience,
not sentience,
artificial sentience
and our artifacts,
well,
this is where
it's like natural consequence.
So,
I like to think of it
as like an extension
in like placing a pathway
if you come from
that line of thinking.

So,
all of you guys,
the Active Inference Institute
is like,
perfect candidates,
right?
Because you're,
you're studying deeply
the free energy principle.
So,
I think mortal computation
is just like
the next step,
right?
It's like,
okay,
well,
that's another piece
to this puzzle.
It's where you go
if you want to build
technological artifacts
with what we would
consider sentience,
if that makes sense.
Lou,
want to add anything?
Sure.
I mean,
I have millions
of questions,
but nothing pressing.
But this one,
I think,
maybe goes back
to you talking
about cognitive diversity
and then Daniel

mentioning like shining
the light of attention.
And just as we're,
as we're thinking
about like attention
as maybe like
the foundation
for cognitive diversity,
because like,
while we are all
fundamentally have
the same programming,
the same like voluntary
and involuntary consciousness,
right?
I guess we'll say it like that,
like my heartbeat
and kind of the,
my self-sustaining system.
we can kind of make that conscious.
Like I've mentioned the breathing,
but also like just through
like motor control,
like you can activate
motor circuits.
Like we all have different,
like some people
can spread their toes out
and some people cannot.
Like if you've never tried
to do that,
like it's hard
and you have to like
practice,
practice,
practice.
But then like
through constantly

paying attention
to those like neural connections,
neuromuscular connections,
you,
you gain like a different
perspective
than others.
And so just
in terms of cognitive diversity
for humans,
it's going to look
really different,
like within
like intraspecies
and interspecies
diversity
is going to be crazy,
right?
Like,
you know,
a tardigrade
doesn't even have a toe
to spread.
Maybe they do.
I don't know.
I don't think they have toes,
but,
but just across
the different species,
maybe we,
this is the time
to kind of look
a little closer
at like procedural memory,
long-term memory,
like at what,
like what is long-term memory,
right?

Like if,
you know,
when you're looking
at something like an insect
that doesn't really have a brain,
definitely doesn't have
like a hippocampus
to store the long-term memory,
do you get
long-term memory?
Does it go into
the generative model?
Like,
or where does it go?
Like,
I mean,
you think about it,
they just have like
the working buffer memory,
or if you were to try
to design
one of these
cognitive kernels
for a non-human,
like maybe
you don't have to go
to a plant,
but like,
where would you
start to kind of
add or subtract
things like long-term memory?
Because we know
that that's really different
just even in the animal kingdom.
Mm-hmm.
Well,
I,

first of all,
I agreed with most
of what you were saying
at the beginning.
That just resonated
with cognitive diversity.
If,
talking about like,
how would I design like,
or begin to think
about something
like a fruit fly
as opposed to,
you know,
a human,
again,
the mortal computation
definition
and like the general framing.
And let's,
just again,
for the audience members
that might join in
and say,
wait,
what are they talking about?
That's not in any
of the four papers
we're mentioning.
Let's just think
of the mortal computation
thesis as subsuming
all of these
because I think
of all those,
just as a quick point too,
all of that work,
I think,

is in service of this.
Even though this was written
far more recently
and I know,
Blue,
you were going to ask
at one time,
what was the history
of the ordering
of these papers?
But obviously,
they're not in a
strict sequence.
So it's like,
mortal computation
is like,
one way to do
a synthesis
of all these ideas.
Now,
to build like something
like a fruit fly,
mortal computation
just says
you need
at least at its heart
a generative model,
right?
Everything's a generative model,
but the form
of that generative model
is not,
you know,
it's not specified
because there's
the principles
like,
for example,

in that paper
brings up
the cybernetic ideas
of like
the good regulator theorem.
I'm sure you guys
have encountered this too
because the Utena,
Arl,
especially in any discussion
with those
with free energy,
they always borrow
from cybernetics.
It's impossible.
So law of requisite variety
and the good regulator theorem
are like kind of like
the most well-known.
That paper talks about
a lot of other ones
that I don't think
get enough attention.
So all the axioms
of cybernetics
are kind of in that paper.
But the good regulator theorem
and the principle
of model internal control,
which is not as popular
as good regulator,
but you sort of need those
at the formation
of any of these entities.
And so as long
as you're able
to design
the body

of that fly
and you are,
I'm assuming
you're going to study
the object in reality,
right?
So you're going to study
the actual fruit fly
and hopefully extract principles
because again,
this is biomimetic.
So the title
of the paper says
we are mimicking
or performing mimicry
of biology
or the things
that we perceive
in our real world,
the physiological organisms.
And so if you're able
to study the fruit fly,
I would imagine
you're constructing a body
that's very similar to it,
tries to adhere
to as many of its physics
and physical principles
that govern how it moves
and interacts
and its exchange
with materials.
If you have all that part,
I think you're almost there.
Believe it or not,
that's the kind of
the fun part
of the mortal computer

is designing
that Markov blanket
and implicitly,
explicitly,
however you want
to go about it
and placing in a niche
gets you really close
because again,
embodiment offloads
a lot of that cognition
because that fruit fly
does a lot of what it does
based on reactions
and its interactions
with its niche.
And that is where
in like the 5e cognitive theory
of the paper,
the bottom,
the most,
let's say,
you need this
because it influences
everything else
is called,
well,
Carl and I did introduce
it's a rewrapping
of basal cognition.
You need elementary cognition,
which you could just,
if you want to call it
basal cognition,
that's also fine.
We cite the heck
out of that
in Michael Evans' group

and many other biologists.
So you would be
implementing that level.
So you'd say,
if I'm going to build
the fruit fly,
I'm not going to be building,
you know,
you know,
embodied,
well,
you're going to get
embodied cognition
from that,
but you're not going to be
building these complex processes.
You're going to focus
on that slice.
That's what's kind of nice
about 5e theory
is you can slice it up
and say,
I'm going to design
at this level.
Obviously,
the bottom level
affects the upper level,
so you sort of do
have to account for that.
And so you would be saying,
well,
what are the basal
cognitive functions
that define
something like
a fruit fly?
What does it do?
What would evolution

have given it?
Because if you're not
going to simulate evolution,
which would give you
those things,
if, you know,
properly done
and with enough compute,
you're going to design
those inductive biases,
you're going to hopefully
study the fruit fly
and then its internal
cognitive model
will not be
the common model
of cognition
because the common model
of cognition
is sort of like,
okay,
that was the product
of the human evolutionary track,
right?
And that's what evolution
gave us
is like our organizing
principle of the brain.
But other animals
don't necessarily have
that to the same degree.
And creatures,
as you get smaller
and smaller
and simple,
let's say,
different on the spectrum,
especially as you move

towards things
that are a little bit simpler
like single cell organisms,
they're going to have
a lot less of that functionality.
But still under the hood
is that generative model.
It just might not be
a generative model
as in like,
oh, I have a neural circuit
that's trying to predict
the next state
of the environment
like a fantasy
or a dream sequence
that I unroll out.
Like we humans
can sort of like,
oh, let me think about
what's going to happen
if I close my eyes
and I knock over the computer.
We can use our
internal physics simulation
to figure out
what it's going to do
and say,
nah, I'm not going to do that.
Whereas a flute fly
might have some sense of physics,
but it might not be like,
oh, I'm rolling out
over a long-term trajectory.
So I'm not going to be building,
for example,
a procedural memory
or an episodic memory

quite like what I do
in the common model
of cognition.
And I might only need
very simple
little neural circuits
because again,
the number of neurons
and for example,
like insects
is really, really, really small.
And that's why
we can study them better too.
That's also why
people study that.
Or even like,
it's not necessarily
the same as a fruit fly.
It's a lot simpler,
but still fascinating
is C. elegans, right?
It's been studied a lot
because we can map out
every single bit
of its neural circuits
and that in and of itself
is rather complex.
So I would argue
that the fruit fly
cognitive model
will be simpler.
And it will be dictated
by what we decide
are the elementary
cognitive functions
that make
its mortal computer
it a mortal computer.

And we won't need,
you know,
these complex modules
interacting in the way
we do
in, for example,
you know,
a human model
or let's say
an animal model.
I think that's,
but again,
that's grounded
in something like
you understand
your mimicking.
So if you go
with the route
of biomimicry,
you should probably
be studying the fruit fly
or at least understand
what we understand
about it
and dissect that
and bring back
what are its base functions
and then how am I
emulating that?
And then,
you know,
you build your little
cognitive internal states,
right,
inside the Markov blanket
that maintain,
for example,
what are the essential variables

that the fruit fly
needs to maintain?
It's not going to be
the same biochemical
processes
that dictate a human,
right?
Like, for example,
when we have blood,
we're looking at like
pH level
and temperature.
The fruit fly
might be looking
at a little bit
different essential variables,
right?
Certain chemical concentrations
matter to it more.
You'd want to model that.
And I think that that,
by the way,
when you talk about
the basal cognition
in designing it,
my thought on this
is that you are
approaching it
from what are those
life-sustaining variables
that characterize that?
Start from there
and then you say,
okay,
well,
does that explain
some of the things
that I observe,

you know,
let's say,
again,
we're modeling
at this point,
but, you know,
what the fruit fly does,
like, for example,
flying around,
why does it fly around?
Well,
maybe it's because
it needs to,
you know,
reach this particular nutrient
and so it knows
if I go to this region,
I am more likely
to achieve it.
That might be
an allostatic process,
right?
And you'd say,
oh, okay,
there's something
that I need.
And this could also be
like a bottom-up design,
right?
If you don't think
that you need,
like,
the inductive bias
that's provided
by the common model,
you sort of build
your little functions
as needed

and now you build
your little internal
generative model,
your good regulator,
right,
into the fruit fly
based on the key functions
it needs to satisfy
so that it does not die,
right?
And ultimately,
the best way
to probably do this
is then you simulate evolution,
right,
and maybe you let
some natural selection happen
because if you don't want
to design those processes,
then you'd want to simulate,
you know,
the origins of it.
And by the way,
mortal computation
doesn't ignore evolution here.
It doesn't talk about designing it
because, you know,
that's a whole realm
in and of itself,
but even the life-mind
continuity thesis
from Kirchhoff's group,
they talk about
you need a selection history
and I think the only way
to get around
the selection history
is then you say,

well, okay,
if I'm not going to simulate
the selection history
and let, you know,
these primitives emerge
into the structures
that make, for example,
fruit fly work,
you need to figure out,
well, what was those,
what was the product
of our current
non-selection history, right?
What did evolution do
to give the fruit fly as is?
So you've got, like,
two choices.
Study the real thing
and try to emulate
some of those
very basic principles,
like identifying
its essential variables,
identifying allostatic processes
that might service
those essential variables
and the homeostasis
of them
or the homeohesis
of them
or talk about
some very basic primitives
and then say,
I'm going to simulate
a population of these
and that might be
more difficult
because if you're

dealing with the actual,
like, let's say
you have a little
fruit fly robot
since we're talking
about substrates,
that might be
not very cheap to do
where it might be easier
to do it in simulation
and doing it
with a digital morphology
and that's another
can of worms.
So I don't know
if that kind of
gives you something blue.
No, it definitely does.
And, like, yeah,
so, sorry,
my brain works
faster than my tongue.
Sometimes insects
definitely have brains,
just nothing like our brains.
They just definitely
don't have brains
like ours, right?
Like, so nothing
that we can map
analogously over,
although some
of the neurotransmitters
and things still work,
but just in a different
functional way
to my knowledge.
Also, Andrew Barron

is building, like,
a tremendous model
of the honeybee brain
that is just so super cool,
I think,
if you don't know his work.
It's really, really neat.
Oh, give me a link to that.
I'd love to see that.
That sounds awesome.
Yeah, it's way cool.
Has him running around
in mazes
and doing all kinds
of learning, like,
how to tell a Van Gogh
apart from a Monet.
I mean, they're really doing
some pretty complex stuff,
and that's why, like,
it, you know,
it always makes me wonder,
like, where does this stuff go?
Like, where is the memory there?
And it's just such a
fascinating process to me
how memories are created
and stored
and, like,
how they map into the brain
and, like, sometimes
I can pull something up
right away
and, like, sometimes
it'll take me four days
to remember someone's name.
Like, what was the name of that?
And, like, I'll remember it.

I'll be just driving
down the road.
Like, it was whatever.
And, like, it's just, like,
how many layers
of circuitry
did I have to, like,
transgress
to get from A to B,
you know?
And, like, yeah,
how do different kinds
of animals do that
is always really fascinating
to me.
I think a piece of that, too,
is evolution
also has a memory, right?
And I think that selection history
is what goes into things
like, for example,
the fruit fly, right?
So there's
its predecessors,
you know,
you know,
maybe they did
certain other behaviors
that were not conducive
to their survival.
I think there's that, too,
that implicit kind of memory.
And then that gets encoded
in the morphology, right?
And this is where
it goes back to
where I just said
if you want it,

like, your collaborator
or your friend
that you mentioned
about designing,
like, the honeybees,
it's kind of the same principles.
You say, well, okay,
what did evolution give it?
Let me design that.
You're just skipping
the evolutionary phase.
And I think there's some memory
in the structure, too.
So this is where
the body...
Now, of course,
with your example
about you retrieving it,
there's layers
of intricacy there.
I'm not complaining about that.
But even, like,
the fruit fly,
why does it remember
to do certain things?
Or why does it seem
to have certain patterns?
And that's also
what the moral computation thesis
does account for,
or at least tries to,
by saying that
it is because
that body changes
and it's, you know,
subject to a selection history,
because it's constantly
in this eternal change

of repair and maintenance
in the face of disintegration,
because it's always
the threat of disintegration.
Your identity will cease
when you die.
And so because of that
things get encoded
in that memory,
and that's also
this beautiful piece
of, like,
the interplay
of physics and cybernetics.
This is without ever
getting into
the probably wonderful
quantum interpretations
I'm sure Chris Fields
could enlighten us all about.
But, you know,
the idea that
information or variety
gets encoded
into these interact...
through its interactions,
gets encoded
into the morphology.
The very organization
and structural manifestation
of that organization
because organization
is not structure.
It is the relationships
of the parts
and it can exist
independent of the manifestation
and the manifestation

is like,
oh, okay,
here's the body
that embodies
those principles.
That also encodes.
That is a memory.
Because, for example,
an organism that understands
a particular chemical interaction
might say,
well, look,
I need to filter out
this toxin
or I'm going to die
and I need to pump
in these nutrients.
That's a memory.
It's just not
a retrievable memory.
It's just like
an implicit memory.
But that's why,
for example,
it's able to conduct
the actions that it does.
And this is why
mortal computation
really tries to...
And this is where
I think it's very
forward-thinking
or very far ahead
of what we can
probably even do easily,
which is,
it really says
you do need

to understand the body.
The substrate
is so important
and understanding
how that will then
affect the internal states
and how you do
this processing.
That has to be
at the front and center
of your story,
of whatever you build.
And then the niche,
you can't escape the niche.
You're kind of,
by having a body,
you're like,
well, okay,
but the body only existed
because it was in response
to the environment
and it was,
you know,
it has evolved
or it has adapted
because of the perturbations
the environment
has subjected it to.
And so I think
that that's why
I like the thesis so much
is it's kind of say
we need to flip
our thinking a bit.
Not, okay,
we're going to design
the brain
and someday in the future

we'll have good bodies
for it.
I think that
that was valuable
and it has produced
some interesting things,
but it can only get you so far.
And then you miss
at least 50% of the puzzle
that could also inform
that design
where we talked earlier
about like emotions
and, you know,
I don't,
I wouldn't argue
that the systems
that we have today
despite the media claims
and, you know,
there's the stuff
with OpenAI
and I saw some comments
about, oh, well,
you know,
Alten and others,
they have some secret
AGI program,
but I've heard conspiracies
about that stuff
all the time.
So I don't,
I put too much credence
to it.
I don't think
that they've cracked
the emotional nut,
the biophysical nut.

I think it's more like
we've had lots of people
either argue,
study very specific
slices of it,
but those voices
need to be echoed,
right?
And so that's what I think
the moral computation
is an echo as well
of at least
like a century of thought
and it's saying
this is how we can approach
artificial general intelligence.
And then it can account
for things
like the fruit fly.
Actually,
you said Andrew
was his name
on the work
on like honeys.
That to me
is an example
of if,
you know,
he's got a learning component
and an inferential component
and there is some organizing,
he has a body,
he has a niche,
you know,
that's very close
to an in silico version
of a mortal computer,
right?

You know,
the only other components
that it might be more
like an allostatic MC.
So,
and again,
in the paper,
we have a table
and Carl did want that
to stay,
I was going to move it
to the appendix again,
length.
So,
we made a compressed table
of different types
of mortal computers
and we kind of characterize them.
There's a few
that are autopedic.
They actually do have very,
like,
organoids
really,
really close
to what you would want,
right?
But they're all biophysical.
So,
there's not like a,
oh,
I designed something
on a neuromorphic chip
with predictive coding,
like what I would do.
This is,
you know,
oh,

look,
we're using stem cells
to construct
this microphysiological
architectures.
I do think
that has a place,
a super important role
going forward
and I would only hope
that there's some
democratization
of an exchange
of ideas
between this wonderful,
really new world
and machine intelligence
to learn
and study
organoids
but also fungal
and mold-based computing.
Really interesting
place to be as well
and to think about
how does that
biophysical organization
give rise
to community
and those are examples
of mortal computers.
Ashby's homeostat
deserves a lot of love
and attention
because it's the oldest
and really the most
prominent example
of what I would argue

is a true in silico
mortal computer
but it's homeostatic.
A little bit allostatic
depending on what you've done.
There's variations to it
but like
the xenobots
really interesting.
I think the xenobots
are an example
of something
that they use
frog cells
like cells
and pieces of frogs
to actually
instantiate
these little tiny
robots
that can self-heal
and repair
and I think
that's going to be
the part where
we have
autopedic computers
already in front of us
like organoids
and xenobots
are some of the most
interesting
I think
profound examples
of how you get
this computer
and then for example
Andrew's work

would have
you know
maybe the missing
component
so his would be
more allostatic
so mid-level
kind of mortal computers
and I'd have to read
more about them
but just from your
explanation
but the structural
selection
the evolution
of the morphology
the self-replication
part of the story
which we also
talk about as well
is more
in like the xenobots
and so I think
it's interesting
how do those
two worlds interact
and how do you get
those properties
of self-repair
self-replication
which by the way
if you add a mutation
to self-replication
you get reproduction
because now
that creates
a new system
that's now subject

to natural selection
and now you've
you know
essentially done reproduction
and now we have
an account
your computer
has all the pieces
that you know
complex organisms
do to also
ensure their persistence
which is
we have offspring
right
you know
that's another way
we project ourselves
into the future
in an indirect way
so it'd be kind of cool
to take like
the honeybee model
and think about
what are the ways
in which we can have
its morphology
be damaged
and conduct self-repair
and that gets into
the real nitty-gritty
and even
it might be difficult
to do right
and that's where
I think
this is where
the grand challenge

right
it's got
an open
pathway
that I don't think
like any one of us
has an exact candidate
you know
for okay
it already exists
because then
there'd be really
no point in writing
the paper
I do think
the xenobots
and organoids
can teach us
a tremendous amount
about
structural selection
morphogenesis
and self-repair
and then
maybe
build like
you know
a synthetic
you know
honeybee
mortal computer
that also
is autopetic
right
it's able to
constantly
repair itself
to some degree

I mean
that would be
very interesting
so
the fact that
you brought that up
was kind of like
oh that's
probably another
example of
an in silico
mortal computer
versus a biophysical
we have
some good examples
of biophysical
not as many
in silico
to my knowledge
even like robots
today
they're not
mortal
right
they
there might be
some interesting
things I'm not
aware of
but
largely
they're like
they have
all the parts
they have
the ingredients
I think
for

mortal computers
but
they're not
they're not
completely there yet
and so
I think
that
making a robot
inherently
whatever the robot
system
it could be
the honeybee
it could be
a creature
that we come up
with
it could be
humanoid
making it
inherently
driven
by just
this fundamental
process of
persistence
is the starting
point
and then thinking
about what are
those essential
variables
and like we
talked about
the fruit fly
designing all
of those

principles
from the ground
up
and then at one
point you've
got to say
is this supposed
to be human
like
am I going to
be able to
evolve
you know
a population
of these
or am I
going to
need to
employ
inductive
biases
like a
common
model of
cognition
and I like
to think
of cognitive
architectures
as a
possible bias
and say
well okay
I'm going to
design a
brain based
on these
principles

it could be
the wrong
bias
but you know
what
we
it's probably
capturing a lot
in you know
cognitive architectures
are fit to a
lot of human
data so
they do
explain a lot
of different
effects
so I
would say
it's a
pretty good
bet
and you
bring in
those biases
and now you
design the
internal states
and now you
have a
very complicated
mortal computer
but you know
the future
will be a
very interesting
place and
hopefully

more interesting
mortal computers
will emerge
in the future
thank you
Alex these
are really
great points
in closing
blue first
as much as
you'd like to
add and
then Alex
any closing
thoughts that
you want to
kind of leave
it with
um just I
don't know
that Andrew
Baron is
building a
robot
B that
would be
super cool
um he's
just modeling
the real
actual biological
honeybee brain
which does
some level
of mortal
computing
I guess

it is a
mortal computer
right like
the the
but it's
not it's
in vivo
not in
silico
um but
Alex thank
you so much
for um
entertaining
all of my
fun questions
and for you
know trancing
us down
this like
super fun
path toward
mortal computing
I'm super
excited to
read your
paper um
and appreciate
your time
on the
stream
yeah
thank you
uh
thank you
blue
and thank
you

daniel
for inviting
me
this was
a really
thrilling
discussion
I really
like your
your guys
podcast
series
and uh
and I
think it's
it's great
that you guys
are pushing
for the
narrative of
free energy
and active
inference
and trying
to educate
the the
greater public
on it
so my
students as
I already
told you
are fans
of your
channel
so I
was uh
it was

my pleasure
to be
on here
um
I hope
you guys
enjoy the
paper
um
and I
guess
as a
one final
comment
I guess
it's now
sort of
our own
private
running
joke
uh
again for
those that
tuned in
hoping to
like have
a deep
dive into
I think
you guys
did a good
job on
point zero
going through
all four of
those papers
to be fair

I think
that captured
their essence
and then
last time
uh
at the
end of
point one
I gave
like a crash
course on
the NGC
predictive
coding
um
so just
to make
a final
comment
to sort
of like
tie
everything
together
to today's
I didn't
actually expect
I
it's my
fault
for introducing
the paper
but it
may it
was a
welcome
uh

derailment
uh
going down
mortal
computer
but I
think
those
four
papers
um
you
could
argue
are like
building
blocks
as well
there are
other pieces
to the
puzzle
of mortal
computation
for example
my
my
preference
or my
thoughts
are
predictive
coding
which is
a
manifestation
of
free

energy
or
variational
free
energy
minimization
that's
going to
be an
important
piece
your
neural
circuitry
should
probably
adhere
to that
because I
think
it's
very
friendly
to a
lot
of
hardware
like
neuromorphic
chips
so that's
kind of
like
where all
the
predictive
coding
stuff

just
to
think
about
my
papers
they're
like
oh
okay
that's
how
that
relates
it's
probably
building
circuits
there
and
building
little
modules
like
we
do
in
cog
engine
the
cog
engine
paper
is
we
have
talked
about

it
even
if
it's
not
about
it
specifically
the
common
model
of
cognition
is
a
wonderful
bias
if you
want
to
skip
ahead
rather
than
evolve
from
the
ground
up
to
human
cognition
and
if
we
want
to
build

a
moral
computer
that
has
aspects
of
human
behavior
we
should
maybe
appeal
to
that
type
of
bias
there's
a lot
more
in the
world
and I
encourage
viewers
to go
into the
world
of
cognitive
science
and
computational
neuroscience
and
dig
into

the
eternal
well
that
is
of
the
knowledge
there
so
I
recommend
that
and
that's
how
that
paper
relates
the
spiking
predictive
coding
and
the
convolutional
which
we
only
touched
on
them
briefly
I'll
just
comment
that
the

convolutional
predictive
coding
paper
which
was
the
second
one
that
you
guys
talked
about
is
like
an
extension
of
the
nature
paper
in
NGC
and
it
just
says
how
do
we
do
vision
processing
I
consider
that
as

another
version
of
the
same
building
block
that
would
go
into
a
mortal
computer
and
the
spiking
predictive
coding
I
think
is
really
interesting
it
says
well
if
I'm
going
to
try
to
get
as
close
as
I

can
to
biophysical
real
neurons
do
so
it's
kind
of
like
we're
building
the
neuron
as
a
little
mortal
computer
and
of
itself
and
we
know
that
they
communicate
with
spikes
and
so
the
spiking
predictive
coding
is

like
one
way
there's
other
ways
to
emulate
predictive
coding
at
a
very
fine
coarse
fine
grained
dynamics
level
and
I
would
imagine
that
would
probably
be
a
great
way
to
build
a
neuromorphic
chip
based
mortal
computer

so
the
idea
is
if
you
get
into
neuromorphics
and
the
mortal
computing
paper
does
talk
about
it
not
in
great
detail
it
talks
about
even
my
other
work
that's
beyond
predictive
coding
like
forward
forward
predictive
forward

forward
only
learning
and
other
variants
and
also
spike
timing
dependent
plasticity
and
how
did
all
those
relate
to
the
mortal
computer
and
that's
in
a
supplement
so
if
you
are
really
a
glutton
for
punishment
and
you

can't
get
enough
there's
an
appendix
with
a
couple
sections
and
it's
talked
about
there
and
so
I
think
that
you
can
just
for
the
viewers
of
this
podcast
this
closing
podcast
you
can
be
like
well
that's

how
it
all
relates
really
it
was
all
parts
of
a
mortal
computer
and
we
can
all
pretend
that
from
the

beginning
I
had
envisioned
a
mortal
just
like
Daniel
brought
up
the
whole
framework
of
active

inference
and
the
free
energy
principle
Bayesianism
they're
all
tools
that
might
help
you
build
you
know
very
powerful
and
interesting
mortal
computers
whether
they are
simulated
or
whether
they're
real
world
and
then
as
we
brought
up
in

this
one
you
guys
have
and
I
think
it's
a
compliment
to
some
of
Chris
Field's
teachings
and
some
of
his
work
there's
also
the
whole
quantum
interpretation
not
quantum
computing
as a
solution
but
rather
a
mortal
computer

and
a
neuromorphic
system
which
that's
what
Carl
and
Chris
did
and
Carl
and
I
did
in
mortal
computation
the
interpretation
from
the
quantum
information
theoretic
could
be
a
very
promising
way
of
formulating
it
and
that
might

open
up
the
doors
to
an
extremely
powerful
form
like
quantum
computing
so
I think
there's
a lot
of
wonderful
wild
out
there
places
to
go
but
they're
not
too
out
there
that's
my
last
point
that
I
want
to

make
they're
out
there
they go
against
a lot
of
our
like
I
said
earlier
held
beliefs
especially
those tied
to
immortal
computation
but I
also
think
that
they
kind
of
build
on
intuitions
and
ideas
that
have
existed
in
other
places

like
if
we
take
biology
and
biochemistry
and
biophysics
seriously
and we
know
that
these
things
work
so
it's
not
like
oh
I'm
going
to
do
something
drastically
strange
and
controversial
it's
just
saying
let's
take
a little
bit
deeper

inspiration
from
the
world
around
us
and
living
entities
to
build
what
could
arguably
be
one
way
to
do
artificial
general
intelligence
and
ultimately
artificial
sentience
which I
think
is
kind
of
the
final
end
goal
right
and
then

yeah
and
we
end
up
there
so
thank
you
guys
again
it
was
so
much
fun
talking
thank
you
for
letting
me
at
some
points
ramble
I hope
my rambling
was useful
but
you know
thank you
I'll give a short
closing remark
so first
thank you
blue for
working on

the dot
zero
diligently
and
thank you
Alex
for like
suggesting
to take
on the
challenge
and
reading
between
and among
the
citations
is like
always
what it's
about
and the
engagement
and the
developments
in the
real
world
so
it's
just
epic
it's
fun
and
it's
lived
the

research
program
is
lived
forward
but
experienced
backwards
or
however
it
was
definitely
very
excited
for
everything
you've
described
and on
all these
topics
I hope
we can
continue
to discuss
and that
people
will be
excited
to
step
up
and
do
like
a
56.3

and we
can just
continue
working
together
in
1790
Immanuel Kant
makes the
famous statement
in his
critique of
judgment
there will
never be
a
Newton
of
the
blade
of
grass
because
human
science
will
never
be
able
to
explain
how
a
living
being
can
originate
from

inanimate
matter
the
German
naturalist
Ernst
Hegel
about
70
years
later
celebrates
Charles
Darwin
to be
such
a
Newton
of
the
grass
blade
Hegel's
enthusiasm
about
Darwin
was not
shared
among
his
contemporaries
and is
not too
widespread
today
although
the
path-breaking

role of
Darwin's
scholarly
work
is not
the least
doubted
or questioned

The American
philosopher,
physicist,
and molecular
biologist
Evelyn Fox

Keller
says that
considering
Darwin
as the
Newton
of
biology
is
simply
wrong
Darwin
himself
has
systematically
avoided
!
natural
selection
begins
with a
living

cell
and I
think it's
actually
this
nexus
that you're
addressing
with the
1900s and
early 2000s
in mind
that is
a big
one
so
thank you
again for
undertaking
the project
and for
joining these
streams
and talk to
you next
time
yeah
thank you
for having
me again
looking forward
to the next
time
bye
bye
to